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Kilogram-Scale One-Pot Synthesis of Multicomponent Fullerene

Composites for Efficient Inverted Perovskite Solar Cells

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Abstract

The high cost and detrimental self-aggregation of conventional single-component fullerenes such as C₆₀ and PCBM constitute a major obstacle to the commercialization of inverted perovskite solar cells (IPSCs). We report a kilogram-scale, one-pot synthesis of a multi-component fullerene composite (FC), comprising C₆₀, a bis((3-methyloxetan-3-yl)methyl) malonate-C₆₀ mono-adduct (BCM), and its bis-adduct (BCB). FC is obtained in 96% yields without complex column chromatography, significantly reducing production costs. Upon thermal annealing, BCM and BCB undergo cross-linking to form a robust encapsulation network that homogeneously incorporates C₆₀, enhancing film stability and

electron mobility. IPSCs incorporating cross-linked fullerene composite (CFC) demonstrate an impressive efficiency of 26.55%, surpassing that of PCBM-based devices (24.82%). Additionally, CFC-based devices maintain 96.0% and 95.1% of

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their initial efficiency after 1,000 hours under ISOS-L-1 and ISOS-D-2 protocols, respectively. Notably, CFC demonstrates excellent performance across a range of device configurations, including wide-bandgap (1.68 eV and 1.77 eV) cells, large-area devices (1 cm²), and mini-modules (14.4 cm²).

Introduction

Perovskite materials have garnered significant attention in the field of solar cells due to their high absorption coefficient,¹⁻³ long carrier lifetimes,^{4,5} and excellent defect tolerance.⁶ Inverted perovskite solar cells (IPSCs) employing fullerene-based electron transport layers (ETLs) offer distinct advantages such as negligible hysteresis,⁷ high power conversion efficiency,⁸⁻¹³ continuously improving long-term operational stability,¹⁴⁻¹⁸ and compatibility with tandem architectures,¹⁹⁻²² making them one of the most promising perovskite photovoltaic technologies for industrialization. A critical challenge limiting their commercial-scale deployment, as universally acknowledged by both academia and industry, lies in the instability of conventional electron transport layers (ETLs).²³ Fullerenes, particularly C₆₀ and (6,6)-phenyl-C₆₁-butyric acid methyl ester (PCBM), have been extensively employed as ETLs in IPSCs due to their excellent electron extraction capabilities and suitable energy level alignment with perovskite absorbers. However, these single-component fullerenes suffer from severe self-aggregation under continuous illumination, which has been conclusively linked to their poor long-term stability and identified as a primary barrier to commercial-scale implementation, as demonstrated by recent collaborative studies between academic laboratories and industrial manufacturers.^{24,25} The aggregation-induced interfacial defects facilitate ion migration from the perovskite layer, accelerating device degradation and compromising operational reliability.²⁶⁻²⁸ Therefore, dense and stable fullerene-based ETLs are essential for ensuring the overall stability of IPSCs.

Cross-linking is widely recognized as a key strategy for designing robust and

stable films across various research fields.^{29,30} In IPSCs, several cross-linkable fullerene derivatives have been reported as ETLs or electron extraction layers to improve device

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stability. For instance, some cross-linkable fullerenes, MPMIC₆₀, [6, 6]-phenyl-C₆₁-butyric styryl dendron ester (PCBSD) and phenyl-C₆₁-butyric acid benzocyclobutene ester (PCBCB), as replacements for PCBM and C₆₀, resulting in improved device stability.³¹⁻³³ However, these cross-linkable fullerenes typically require high temperatures of at least 150 °C to form cross-linked films, which can damage the perovskite (PVSK) layer, limiting their application in IPSCs. Our recent study addressed this issue by discovering that C₆₀ functionalized with bis(3-methyloxazolidin-3-yl)methyl malonate can undergo cross-linking at a significantly lower temperature of 100 °C.³⁴ This not only prevents damage to the PVSK layer but also enables functional groups to chemically interact with the PVSK, further enhancing device stability. Integrating these multiple functions into a single molecule typically requires sophisticated functional-group design, laborious purification steps, and high synthesis costs, making such an “all-in-one” architecture considerably challenging. Meanwhile, researchers have also explored blending various fullerene derivatives to cooperatively enhance performance.^{35,36} However, the synthesis and purification costs of the relevant derivatives remain prohibitively high, which hinders their potential for scalable application.

In this study, we move beyond the pursuit of a single “perfect molecule” and instead develop a fullerene composite (FC) prepared via a one-pot, kilogram-scale process for use as the ETL in IPSCs. The one-pot synthesis eliminates the need for costly separation and purification steps, delivers a yield as high as 96%, and significantly reduces material production costs. Unlike conventional single-component fullerenes, FC consists of C₆₀, bis((3-methyloxetan-3-yl)methyl)malonate-C₆₀ monoadduct (BCM), and bis((3-methyloxetan-3-yl)methyl)malonate-C₆₀ bisadduct (BCB). Thermal annealing at 100 °C triggers cross-linking of BCM and BCB to form a robust network, while C₆₀ is confined within this network. The three components act synergistically, enabling efficient electron transport while markedly enhancing the

long-term stability of the film. The resulting cross-linked multi-component fullerene composite (CFC) exhibits favorable energy level alignment with perovskite absorbers, high electron mobility, and exceptional resistance to ion migration. IPSCs incorporating

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CFC ETL yield an impressive efficiency of 26.55%, surpassing that of PCBM-based devices (24.82%). Notably, CFC-based devices maintain 96.0% and 95.1% of their initial efficiency after 1,000 hours under ISOS-L-1 and ISOS-D-2 protocols, respectively, showcasing their exceptional operational stability. Furthermore, CFC-based ETLs demonstrate high efficiency and long-term operational stability across large-area devices (1 cm²), 14.4 cm² mini-modules, and wide-bandgap devices (1.68 eV and 1.77 eV). This work highlights FC as a scalable and commercially viable ETL, advancing the industrialization of high-performance perovskite photovoltaics.

Results

Kilogram-Scale Fullerene Composite Synthesis

The FC was synthesized via a Bingel reaction between bis((3-methyloxetan-3-yl)methyl) malonate (BOM) and C₆₀ (Fig. 1a). By adjusting the reaction time, three different FCs with varying component ratios were obtained (FC1, FC, and FC3). Specifically, the reaction times for FC1, FC, and FC3 were 10 hours, 20 hours, and 30 hours, respectively. Using our custom-designed scaled-up reaction apparatus, kilogram-scale fullerene products can be readily obtained (Fig. 1b). This result highlights the excellent scalability of the reaction and underscores its strong potential for practical and industrial applications. The presence of C₆₀, BCM, and BCB components was confirmed using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS) (Supplementary Fig. 1). The molar ratios of the components are as follows: for FC1, the proportions of C₆₀, BCM, and BCB are 44%, 39%, and 17%, respectively; for FC, the proportions are 27%, 50%, and 23%; and in FC3, the proportions are 21%, 41%, and 38% (Fig. 1c). The reproducibility of the ternary composition was confirmed by HPLC (Supplementary Fig. 2). The yields of the FCs obtained at different reaction times are provided in Supplementary Table 1. The FC achieved a high yield of up to 96%, and a commercial-scale cost analysis indicates

that the synthesis cost of FC is significantly lower than that of the widely used commercial PCBM (Supplementary Tables 2 and 3). We then fabricated IPSC (device

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structure: ITO/HTL/PVSK/ETL/BCP/Ag, Supplementary Fig. 3) using these various ETLs. As depicted in Fig. 1d and summarized in Supplementary Table 4, devices incorporating CFC ETL exhibited superior efficiencies compared to those using PCBM and binary mixtures of FC (Supplementary Fig. 4). Statistical data on device performance (Supplementary Fig. 5) indicate that CFC-based devices show significant improvements in open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), and fill factor (FF) relative to PCBM-based devices. Notably, the device using CFC achieved an efficiency of 26.55%, surpassing the performance of devices with PCBM (24.82%) and those with CFC1 and CFC3 (Fig. 1e). Furthermore, the forward and reverse scan data demonstrate that IPSCs based on CFC exhibit reduced hysteresis compared to those based on PCBM (Supplementary Fig. 6 and Supplementary Table 5). The device performance also demonstrates high robustness against variations in the CFC layer thickness (Supplementary Fig. 7).

We also evaluated the stability of the devices under maximum power point (MPP) tracking conditions with one sun illumination. As shown in Fig. 1f, the unencapsulated device based on CFC maintained 96.0% of its initial efficiency after 1,000 hours under the ISOS-L-1 protocol, whereas the PCBM-based device preserved only 47.9% of its initial efficiency. Additionally, we assessed the thermal stability of the devices. The CFC-based device retained 95.1% of its initial efficiency after 1,000 hours of aging under the ISOS-D-2 protocol at 85 °C, while the PCBM-based device maintained only 66.7% of its initial efficiency (Fig. 1g). Meanwhile, the CFC-based devices demonstrated superior ambient storage stability compared to the PCBM-based devices. (Supplementary Fig. 8).

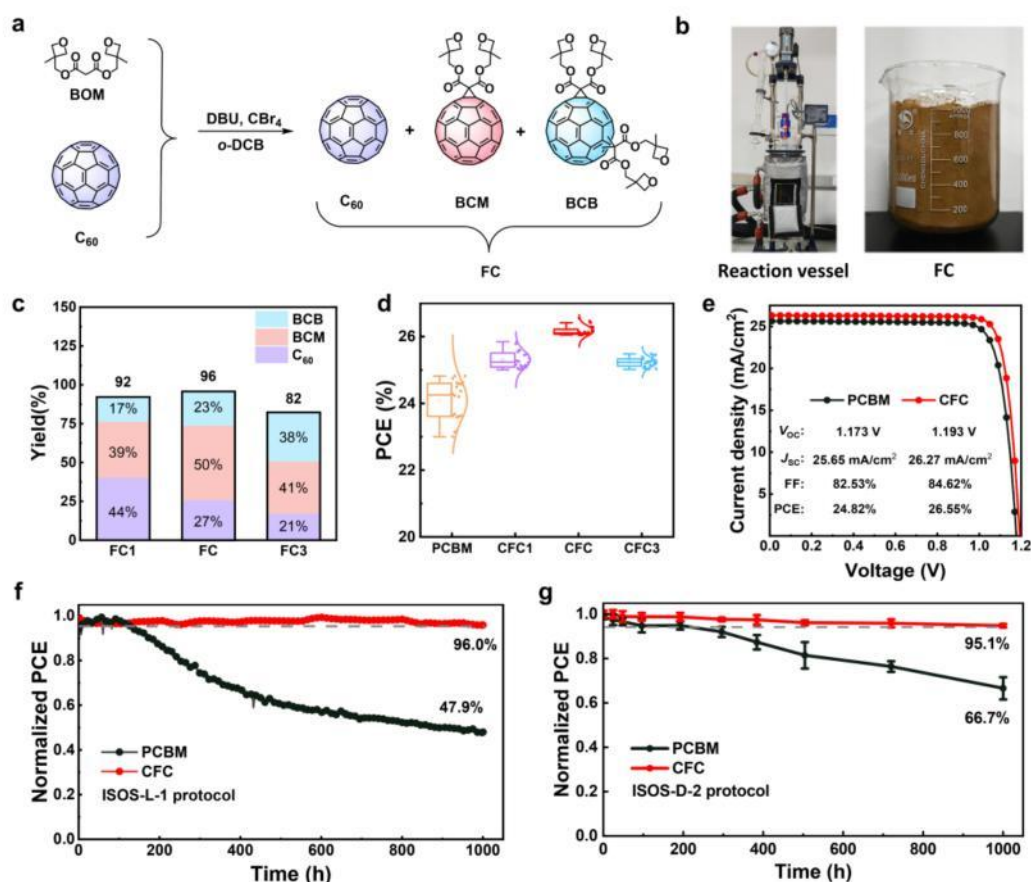


Fig. 1 | Kilogram-Scale Fullerene Composite Synthesis. **a**, Synthetic route of FC. **b**, Scaled-up reaction vessel and kilogram-scale product. **c**, Yield of FCs at different reaction times and the composition percentage of each component. **d**, Statistical analysis of device efficiency with different ETLs, the boxes show the 25th and 75th percentiles, and the whiskers display the 5th and 95th percentiles. The median and mean are indicated by the dividing lines across the boxes and the open square symbols, respectively. **e**, $J-V$ curves of the devices with different ETLs. **f**, Device stability under one sun illumination according to the ISOS-L-1 protocol. **g**, Device stability at 85 °C for PCBM- and CFC-based devices under the ISOS-D-2 protocol, and the data is plotted with standard deviations (SD).

Film Characteristics of the Fullerene Composite

To understand the reasons behind the changes in device stability, we employed time-of-flight secondary ion mass spectrometry (ToF-SIMS) to track the distribution of Ag^+ , Ag_2I^+ , C_{60}^+ , Pb^+ , and FA^+ ions in PCBM- and CFC-based devices before and after 1,000

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hours of MPP tracking. As shown in Fig. 2a and Supplementary Fig. 9, the electrodes of the PCBM-based devices exhibited significant darkening, indicating severe corrosion. A large quantity of Ag^+ ions was detected in the PCBM and PVSK layers, and FA^+ , Pb^+ , and Ag_2I^+ species were specifically identified in the silver electrode. This suggests that the PCBM ETL is unstable under long-term operational conditions, allowing ions such as FA^+ , Pb^{2+} , and I^- from the PVSK to penetrate the PCBM layer and reach the silver electrode, resulting in electrode corrosion and PVSK degradation. Cross-sectional scanning electron microscope (SEM) images of the device after 1,000 hours of MPP tracking (Supplementary Fig. 10) showed significant aggregation in the PCBM film and noticeable changes in the PVSK phase. X-ray diffraction (XRD) patterns (Supplementary Fig. 11) presented a significant increase in the PbI_2 peak in the PVSK film with PCBM as the ETL, indicating substantial degradation of the PVSK. In contrast, in CFC-based devices, FA^+ and Pb^+ ions were mainly accumulated in the PVSK layer, with no apparent corrosion observed on the electrodes (Fig. 2b). This observation suggests that the CFC layer effectively blocks ion diffusion from the PVSK layer to the silver electrode, reducing electrode corrosion and PVSK degradation. Additionally, cross-sectional SEM images showed that the CFC film maintained its intact structure, and the PVSK layer preserved its morphology throughout the same MPP tracking period, with no significant changes observed in the XRD patterns of the PVSK phase. Based on these findings, we attribute the improved device stability to the compact and stable CFC-based ETL, which effectively eliminates molecular aggregation and reduces ion diffusion from the PVSK to the silver electrode.

To further elucidate the mechanism underlying the enhanced device stability, we prepared FC film from chlorobenzene solution and found that, after heating at 100 °C (Fig. 2c), it became insoluble in chlorobenzene, indicating that FC undergoes cross-linking at this temperature. The results from annealing-time-dependence and solvent-resistance tests on the CFC film (Supplementary Figs. 12 and 13) reveal that an 8-

minute annealing step is sufficient to induce the formation of a stable crosslinked fullerene film. X-ray photoelectron spectroscopy (XPS) revealed that the C-O peak shifted from 533.9 eV before annealing to 534.1 eV after annealing (Supplementary Fig.

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14), consistent with oxetane-ring opening and cross-link formation. Fourier-transform infrared spectroscopy (FTIR) further supported this conclusion, as the characteristic oxetane absorption peak at 983 cm^{-1} disappeared after annealing (Supplementary Fig. 15). In accelerated aging tests involving continuous annealing at $100\text{ }^{\circ}\text{C}$, optical microscopy images (Fig. 2d and 2e) revealed pronounced granular aggregation in PCBM films after 200 hours, indicating significant self-aggregation. In contrast, CFC films showed no noticeable changes before and after aging, confirming their superior thermal stability.

The compactness of the functional layer is crucial for the stability of PVSK devices. To evaluate this, we measured the densities of CFC and PCBM films using the Beer-Lambert law. First, we measured the absorbance of FC and PCBM films at different concentrations (Supplementary Fig. 16) and established the relationship between absorbance and concentration (Supplementary Fig. 17). Using the UV-visible (UV-Vis) spectra of dissolved films solution and the thickness variation from FC to CFC films (Supplementary Fig. 18), the densities of CFC and PCBM films were calculated to be 1.0670 g/cm^3 and 0.8427 g/cm^3 , respectively (Fig. 2f). Additionally, we monitored the changes in water contact angle over time for different fullerene ETLs (Supplementary Fig. 19), revealing that CFC-based films exhibited superior hydrophobicity and stability. These findings suggest that dense and stable CFC films not only protect PVSK from water and oxygen erosion but also effectively prevent ion diffusion from the PVSK to the metal electrode, thereby enhancing device stability.

Building on the exceptional properties of CFC, we propose a schematic diagram to illustrate its functional mechanism (Fig. 2g). The cross-linkable fullerene components within CFC (BCM and BCB) undergo synergistic cross-linking to form a robust network that effectively encapsulates and stabilizes C_{60} . This strategy enables the formation of a low-cost, high-density fullerene composite ETL with improved electron mobility and stability, thereby promoting the fabrication of highly efficient and

stable IPSCs.

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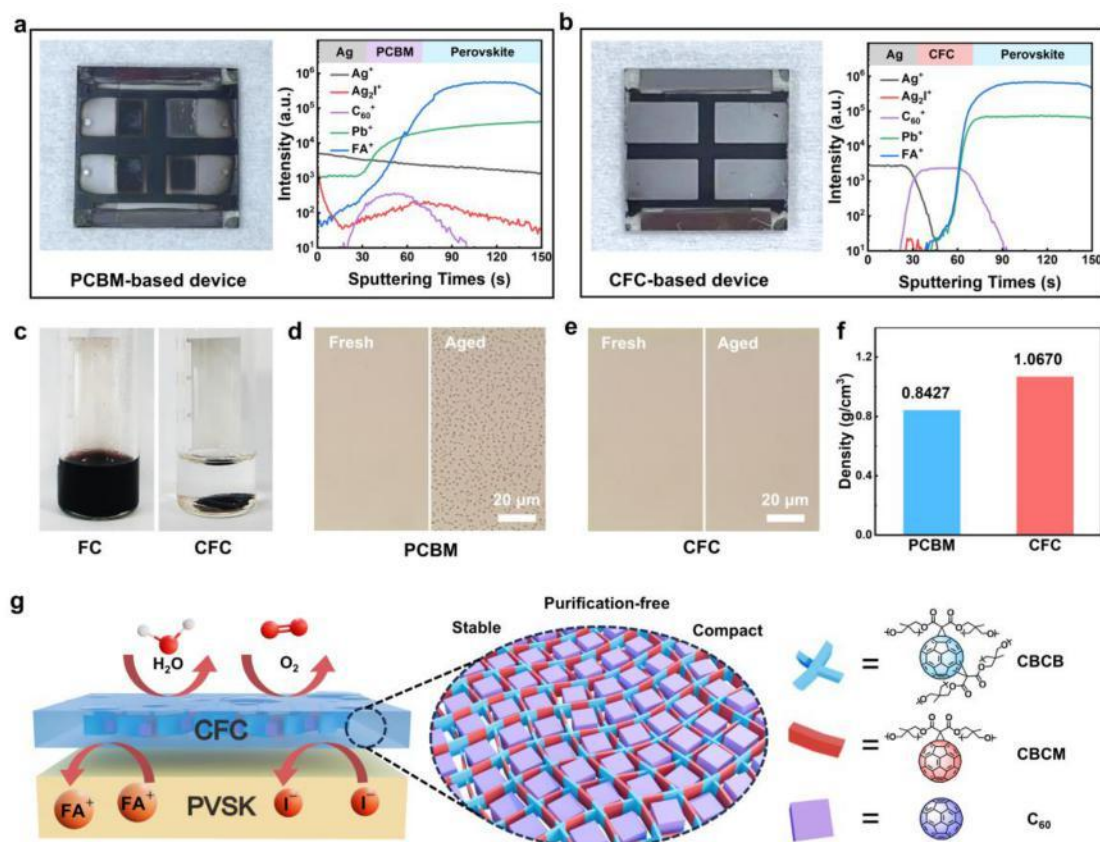


Fig. 2 | Film Characteristics of the Fullerene Composite. **a, b**, Photographs of the electrodes and TOF-SIMS measurements of PCBM- and CFC-based devices after 1000 h of MPP tracking. **c**, Images of FC and CFC in chlorobenzene. **d, e**, Polarized optical microscope images of PCBM and CFC films on glass before and after 200 hours of thermal aging at 100 °C. **f**, Film density comparison between PCBM- and CFC-based ETLs. **g**, Schematic illustration of the CFC ETL, which is formed via the synergistic combination of C₆₀ and cross-linkable fullerenes.

Enhanced Charge Transport in the Fullerene Composite

Proper energy level alignment with PVSFs and high electron mobility are also crucial for fullerene-based ETLs. We characterized the band structure of fullerene films using ultraviolet photoelectron spectroscopy (UPS) and UV-Vis absorption spectroscopy. As

shown in Fig. 3a, the valence band maxima (E_{VBM}) for CFC and PCBM films were determined to be -5.65 eV, and -5.66 eV, respectively. Combining these results with UV-Vis absorption spectra (Supplementary Fig. 20), the conduction band minima (E_{CBM})

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for CFC and PCBM films were calculated to be -3.90 eV and -3.93 eV, respectively (Fig. 3b and Supplementary Table 6). These results indicate favorable energy-level alignment between CFC and the perovskite absorber, as further confirmed by KPFM measurements (Supplementary Fig. 21). We also fabricated electron-only devices with an ITO/Al/Fullerene/Al structure and recorded the I - V curves under dark conditions. As shown in Fig. 3c, the electron mobility of the CFC film was calculated to be $4.02 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, which is higher than that of PCBM ($1.76 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$). This indicates that the CFC film demonstrates superior electron transport capability compared to PCBM films.

The interfacial interactions between the ETL and PVSK facilitate efficient carrier extraction and defect passivation.^{37,38} In the X-ray photoelectron spectroscopy (XPS) of PVSK/CFC films (Fig. 3d), the Pb 4f peak shifts towards lower binding energies, suggesting a coordination interaction between the C=O groups in CFC and Pb^{2+} . Further confirmation of this interaction is provided by FTIR measurements, where the C=O stretching peak of CFC shifts from 1749.6 cm^{-1} to 1738.5 cm^{-1} upon mixing with PbI_2 (Supplementary Fig. 22), in agreement with the XPS results (Supplementary Fig. 23). Theoretical calculations indicate that the PVSK/CFC interface exhibits stronger interactions and more effective defect passivation compared to PVSK/PCBM. (Supplementary Fig. 24). We therefore evaluated the trap density using the SCLC method. As shown in Fig. 3e, the trap-filled limit voltages (V_{TFL}) of the devices based on PCBM and CFC were 0.191 V and 0.098 V, respectively, corresponding to trap densities of $9.90 \times 10^{14} \text{ cm}^{-3}$ and $5.08 \times 10^{14} \text{ cm}^{-3}$ (detailed calculations in Supplementary Note 2). The significant reduction in trap density indicates that CFC can effectively passivate surface defects in PVSK films.

The influence of charge transfer and recombination dynamics at the PVSK/ETL interface was further investigated through time-resolved photoluminescence (TRPL) and steady-state photoluminescence (PL) measurements. Fig. 3f shows the TRPL

spectra of PVSK films and PVSK films covered with different ETLs. The data, fitted with a biexponential function, reveal average PL lifetimes for PVSK, PVSK/PCBM, and PVSK/CFC films of 1003.11 ns, 30.06 ns, and 24.99 ns, respectively

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(Supplementary Table 7). The significant reduction in PL lifetime for PVSK films with an ETL suggests rapid electron extraction at the PVSK/ETL interface. Furthermore, the shorter PL lifetime for CFC compared to PCBM indicates a higher electron extraction efficiency. Additionally, the τ_2 value for PVSK/CFC (34.12 ns) is greater than that of PVSK/PCBM (31.35 ns), which suggests a reduced density of defect states, in agreement with the trap density measurements. The PL results also show that PVSK films coated with CFC exhibit the lowest PL intensity, further confirming the effective electron extraction capability of CFC (Supplementary Fig. 25).

Moreover, transient photovoltage (TPV) and transient photocurrent (TPC) measurements were performed to study charge transfer and recombination mechanisms in the corresponding devices. As shown in Fig. 3g and Supplementary Table 8, the TPV decay lifetime for CFC-based device is 836.35 μ s, which is longer than that of PCBM-based device (623.33 μ s). Meanwhile, the TPC decay lifetimes for devices based on PCBM and CFC are 14.88 μ s and 4.54 μ s, respectively (Supplementary Fig. 26, Supplementary Table 9). The significantly shorter lifetime of the CFC-based device corresponds to faster charge extraction and transport, which accounts for its higher J_{sc} . Fig. 3h presents the Mott-Schottky plots for devices based on PCBM and CFC. The built-in potential (V_{bi}) for PCBM-based devices is 0.957 V, while that for CFC-based devices is 0.986 V. The higher V_{bi} in CFC devices facilitates carrier extraction and transfer within the device, which helps explain the higher V_{oc} observed in CFC-based devices. Additionally, the ideality factor calculated from V_{oc} as a function of different incident light intensities decreases from 1.79 kT/q in PCBM-based devices to 1.48 kT/q in CFC-based devices (Supplementary Fig. 27), indicating effective suppression of defect-induced non-radiative recombination. Electrochemical impedance spectroscopy (EIS) results (Supplementary Fig. 28) also show a larger charge recombination resistance (R_{rec}) in CFC-based devices, further confirming the effective suppression of non-radiative recombination.

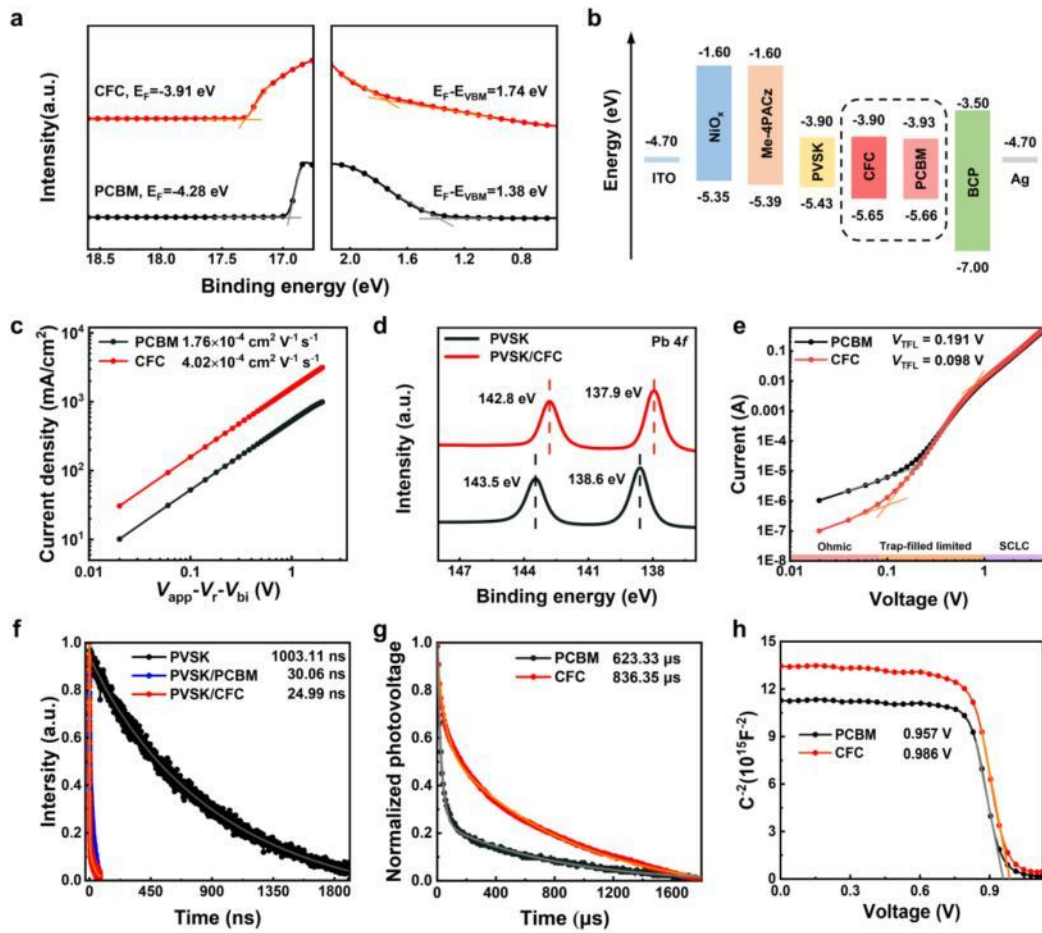


Fig. 3 | Enhanced Charge Transport in the Fullerene Composite. **a**, UPS spectra of CFC and PCBM. **b**, Energy level diagram of NiO_x, Me-4PACz, PVSK, CFC, PCBM and BCP. **c**, Electron mobility measurement of PCBM- and CFC-based films. **d**, XPS spectra of Pb 4f for PVSK and CFC/PVSK. **e**, I - V curves of the electron-only device with the structure of ITO/SnO₂/perovskite/ETL/Ag. **f**, TRPL spectra of PVSK and PVSK coated with different ETLs. **g**, Normalized TPV curve and **h**, Mott-Schottky plots of different IPSCs.

Wide Bandgap Inverted Perovskite Solar Cell Performance

To investigate the applicability of the CFC, we fabricated IPSCs with a 1.68 eV bandgap PVSK absorber, using an ITO/HTL/PVSK/CFC/BCP/Ag structure. As shown in Fig.

4a, the CFC-based 1.68 eV bandgap IPSC achieved a champion efficiency of 23.16%, featuring a V_{OC} of 1.263 V, a J_{SC} of 21.81 mA/cm², and a fill factor (FF) of 84.08% (Supplementary Fig. 29, Supplementary Table 10). In comparison, the PCBM-based

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device exhibited a significantly lower efficiency of 21.21%. Additionally, the CFC-based device exhibits reduced hysteresis (Fig. 4b), achieving a steady-state efficiency of 23.14% (Fig. 4c). We further fabricated IPSCs using a 1.77 eV bandgap PVSK absorber (Fig. 4d). Compared with the PCBM-based device, which achieves a PCE of 18.77%, the CFC-based device demonstrated notable enhancements in V_{oc} , J_{sc} , and FF (Supplementary Fig. 30, Supplementary Table 11), resulting in a PCE of 20.39% with negligible hysteresis (Fig. 4e) and a steady-state PCE of 20.37% (Fig. 4f). The external quantum efficiency (EQE) spectra of the devices are shown in Supplementary Fig. 31, and the integrated J_{sc} values are in good agreement with those derived from the $J-V$ measurements. Their efficiencies are comparable to those of the high-performance 1.68 eV and 1.77 eV bandgap IPSCs currently reported (Supplementary Tables 12 and 13). The stability of the device with a 1.68 eV bandgap PVSK absorber under the ISOS-L-1 protocol is depicted in Fig. 44g. The unencapsulated CFC-based device retains 95.1% of its initial efficiency after 1,000 hours. Similarly, the device with a 1.77 eV bandgap PVSK absorber maintains 95.4% of its initial efficiency under the same conditions (Fig. 4h). Notably, under the ISOS-L-1 protocol, the CFC-based devices outperform their PCBM-based counterparts, suggesting that CFC not only improves efficiency but also enhances the operational stability of IPSCs across different bandgaps. These results demonstrate the versatility of CFC across perovskite solar cells with diverse compositions and suggest its relevance for scalable device fabrication.

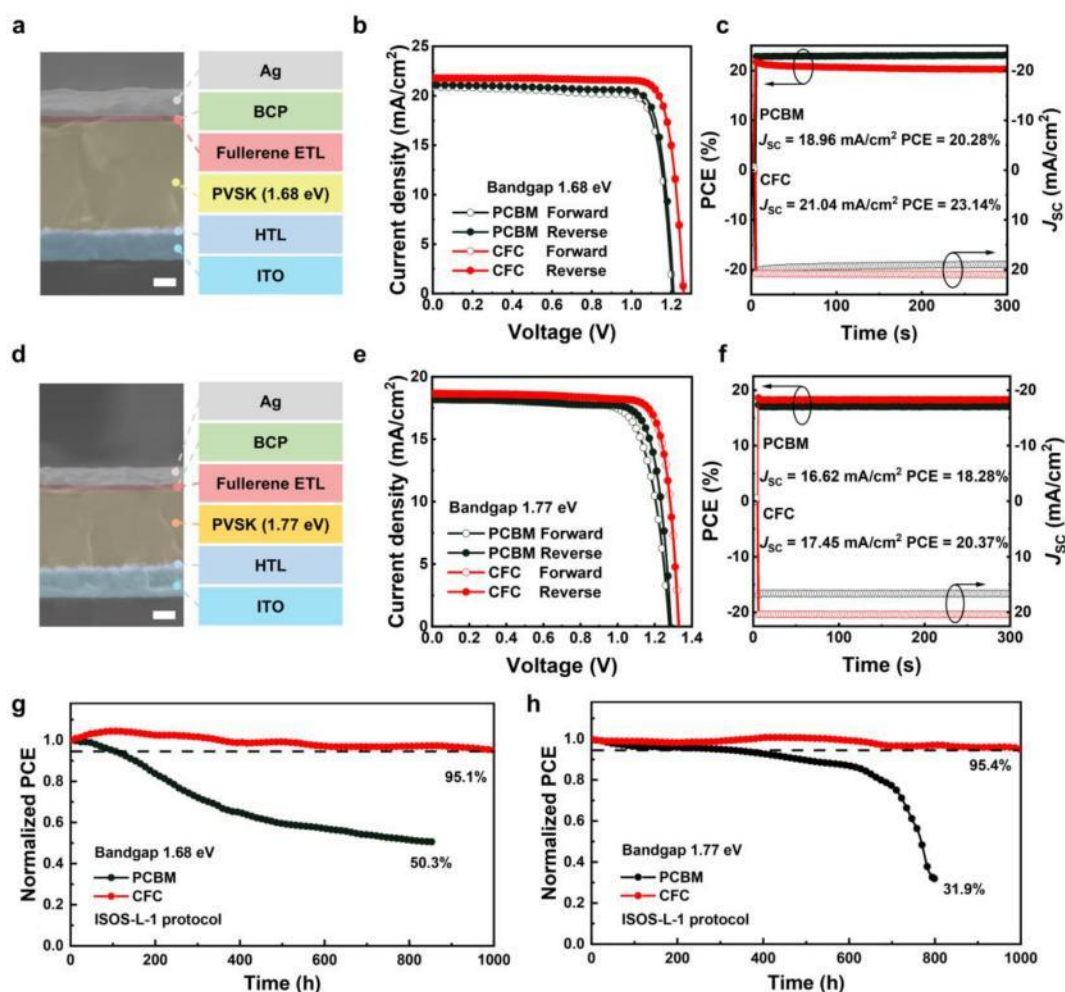


Fig. 4 | Wide Bandgap Inverted Perovskite Solar Cell Performance. **a**, Device structure of 1.68 eV bandgap IPSCs, (scale bar: 100 nm). **b**, **c**, J - V curves and Steady-state output efficiency of 1.68 eV bandgap IPSCs with different ETLs. **d**, Device structure of 1.77 eV bandgap IPSCs, (scale bar: 100 nm). **e**, **f**, J - V curves and Steady-state output efficiency of 1.77 eV bandgap IPSCs with different ETLs. **g**, **h**, ISOS-L-1 stability of 1.68 eV and 1.77 eV bandgap IPSCs under one sun illumination.

Scalable Photovoltaic Performance of Devices and Mini-Modules

To assess the performance of CFC across different scales, we fabricated both 1 cm^2 single cells and 14.4 cm^2 mini-modules. At the 1 cm^2 scale, the champion CFC-based

device achieved an efficiency of 25.36%, with a V_{oc} of 1.191 V, a J_{sc} of 26.02 mA cm⁻², and an FF of 81.85%, outperforming the PCBM-based device (24.08%) (Fig. 5a). The advantage is further confirmed by the steady-state output efficiencies of 25.13% for

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CFC-based device versus 23.96% for PCBM-based device (Fig. 5b and Supplementary Table 14). Additionally, the CFC devices also demonstrated superior device-to-device reproducibility (Fig. 5c and Supplementary Fig. 32).

For the 14.4 cm² mini-modules, CFC exhibited significant advantages. The champion CFC-based mini-module reached an efficiency of 23.81%, higher than the 22.17% of the PCBM-based counterpart, and showed a negligible hysteresis (Fig. 5d and Supplementary Table 15). This trend was further validated by the steady-state output efficiency measured near the MPP. The CFC-based mini-module stabilized at 23.19% over 120 seconds, compared to 22.18% for the PCBM-based device (Fig. 5e), which is consistent with the J - V scan results. Moreover, the CFC-based mini-modules displayed a narrower PCE distribution, indicating better performance uniformity after device scaling (Fig. 5f). The corresponding distributions of V_{oc} , J_{sc} , and FF are provided in Supplementary Fig. 33.

We further assessed the long-term operational stability of both 1 cm² devices and 14.4 cm² mini-modules through MPP tracking. The 1 cm² CFC-based device retained 92.8% of its initial efficiency after 1,000 hours (Fig. 5g), while the PCBM-based device degraded to 77.9% within 400 hours. Similarly, the 14.4 cm² CFC-based mini-module maintained 91.4% of its initial efficiency after 600 hours (Fig. 5h), whereas the PCBM-based device rapidly declined to 25.5% within just 200 hours. These results collectively demonstrate that the stability advantage of CFC is preserved under both area upscaling and prolonged operational conditions.

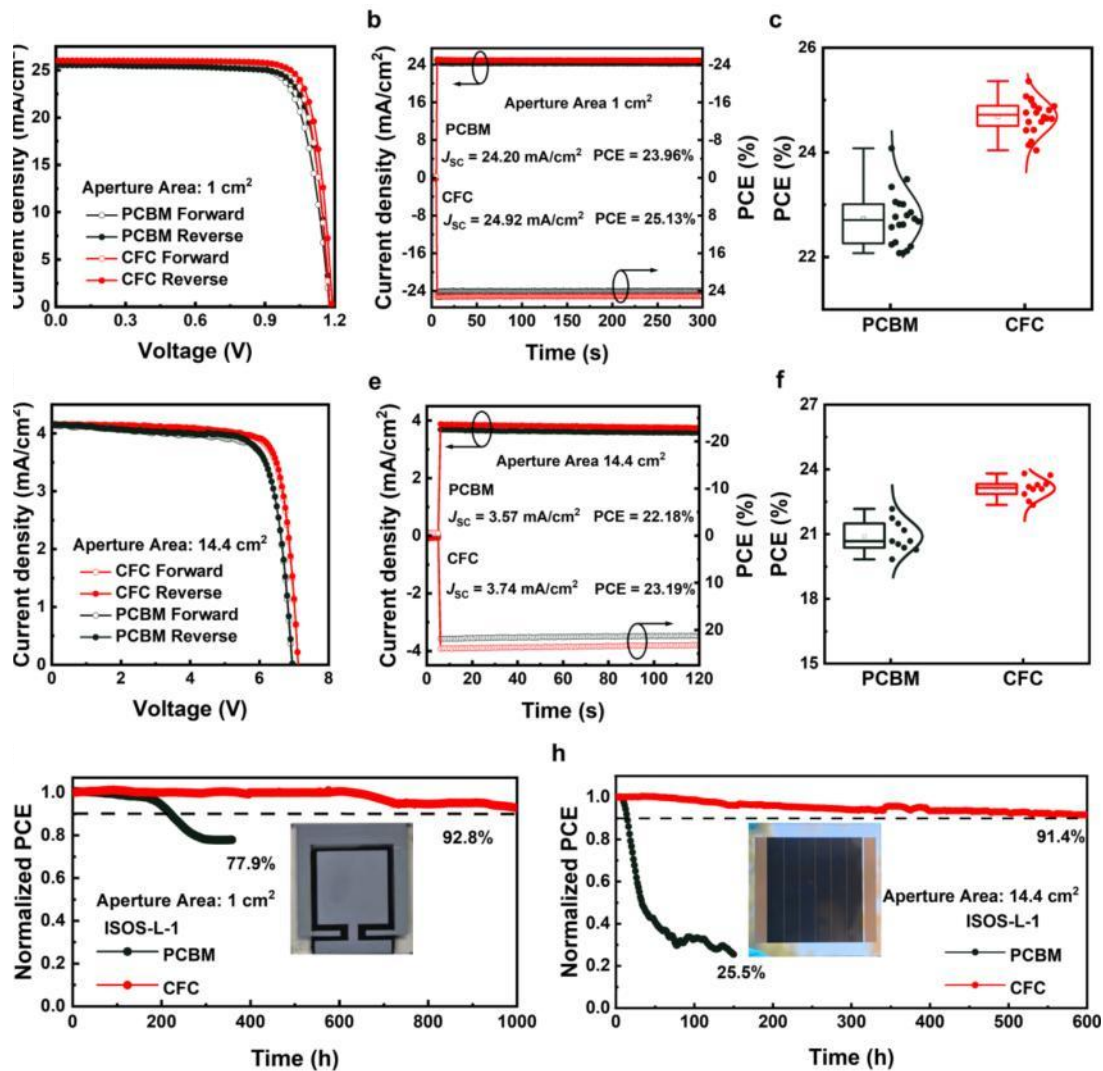


Fig. 5 | Scalable Photovoltaic Performance of Devices and Mini Modules. **a-c**, Performance of 1 cm² devices. **a**, forward/reverse $J-V$ curves; **b**, steady-state output at the MPP; **c**, PCE statistics. **d-f**, Performance of 14.4 cm² mini-modules. **d**, forward/reverse $J-V$ curves; **e**, steady-state output at the MPP; **f**, PCE statistics. In **c** and **f**, the boxes show the 25th and 75th percentiles, and the whiskers display the 5th and 95th percentiles. The median and mean are indicated by the dividing lines across the boxes and the open square symbols, respectively. **g, h**, Photographs of the 1 cm² device and the 14.4 cm² mini-module, and their operational stability under continuous MPP tracking.

Discussion

We introduce a paradigm shift from the pursuit of a single “perfect” molecular material

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toward a rationally designed multicomponent, functionally cooperative system. Through a facile, scalable one-pot synthesis, we construct a ternary fullerene composite (FC) comprising C₆₀, mono-adduct, and bis-adduct fullerenes, wherein each component plays a distinct yet synergistic role. The resulting CFC film, formed via low-temperature cross-linking at 100 °C, enables a champion efficiency of 26.55% in IPSCs, outperforming conventional PCBM-based devices (24.82%). Remarkably, CFC-based devices exhibit excellent operational stability, retaining 96.0% (ISOS-L-1) and 95.1% (ISOS-D-2) of their initial efficiency after 1,000 h. These improvements are attributed to a homogeneous, immobilized C₆₀ network that ensures efficient electron transport while mitigating perovskite degradation under prolonged operation. More importantly, CFC demonstrates remarkable processing robustness and material universality, as evidenced by the performance advantage upon scaling to 14.4 cm² mini-modules and its effective extension to wide-bandgap perovskites (1.68 and 1.77 eV), highlighting its potential as an efficient and stable ETL for inverted perovskite photovoltaics.

Methods

Materials. N, N-dimethylformamide (DMF, 99.8%), dimethyl sulfoxide (DMSO, 99.9%), 2-propanol (IPA, anhydrous, 99.5%), chlorobenzene (anhydrous, 99.8%), anisole (anhydrous, 99.7%), and cesium iodide (CsI, 99.9%) were purchased from Sigma Aldrich. Lead iodide (PbI₂, 98%) and [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz, >99.0%) were purchased from TCI. Methylammonium iodide (MAI, 98.0%) and formamidinium iodide (FAI, 99.99%) were purchased from Greatcell Solar. Lead chloride (PbCl₂, 99.9%) and methylammonium chloride (MACl, 95%) were purchased from Xi'an Polymer Light Technology Corp. C₆₀ (99.5%) and (6,6)-phenyl-C₆₁ butyric acid methyl ester (PCBM, 99.5%) were obtained from Liantan Technology Co., Ltd. Nickel oxide (NiO_x, 99.999%) and Bathocuproine (BCP, 99%) was purchased from Advanced Electronic Technology.

Perovskite precursor solution. 1.53 eV bandgap perovskite precursor solution was

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prepared by mixing FAI, MAI, CsI, and PbI₂ in anhydrous DMF: DMSO=4: 1 (v: v) according to the stoichiometric formula of 1.6 M Cs_{0.05}(FA_{0.98}MA_{0.02})_{0.95}PbI₃, with 20 mol% MAI. The 1.3 M FA_{0.8}Cs_{0.2}Pb(I_{0.8}Br_{0.2})₃ 1.68 eV bandgap perovskite precursor was prepared by dissolving 178.88 mg of FAI, 419.51 mg of PbI₂, 143.13 mg of PbBr₂, and 67.6 mg of CsI in 1 mL mixed solvent of DMF and DMSO with a volume ratio of 3:1. For 1.77 eV bandgap perovskite precursor solution, FAI:CsI:PbBr₂:PbI₂ (molar ratio=0.8:0.2:0.4:0.6) were dissolved in mixed DMF and DMSO solvents (volume ratio=1:1.4) at a concentration of 1.2 M.

Device fabrication. The pre-patterned ITO glass substrates underwent a thorough cleaning process including sonication with a cleaning agent, deionized water, IPA, and ethanol, followed by drying with high-pressure nitrogen and plasma treatment. Next, the NiO_x HTL solution was applied to the substrate via spin-coating and annealed at 100 °C for 10 min. Following this, the Me-4PACz self-assembled molecular solution was applied to create a double-layered hole-transport layer. For the hole transport layer (HTL) solution, 10 mg/mL NiO_x was dissolved in deionized water. After shaking for 20 minutes, the solution was filtered through a 0.22 µm polytetrafluoroethylene membrane before use. The self-assembled molecular layer solution was prepared by dissolving 0.5 mg/mL Me-4PACz in ethanol.

Subsequently, 80 µL of perovskite precursor solution was dropped onto substrates and spun at 2000 rpm for 25 seconds and 5000 rpm for 30 seconds. 10 seconds before the end of the program, 150 µL of anisole was poured onto the substrates. The substrates were then transferred to an annealing stage at 100 °C for 30 minutes. After cooling to room temperature, a PDAI₂ isopropanol solution (1 mg/mL) was spin-coated onto the perovskite films at 5000 rpm for 25 seconds, followed by annealing at 100 °C for 5 minutes. Following this, the FC chlorobenzene solution (20 mg/mL) was spin-coated at 2500 rpm for 30 seconds on the perovskite films and annealed at 100 °C for 10 minutes. Next, the BCP 2-propanol solution (0.5 mg/mL) was spin-coated at 5000 rpm for 30 seconds and annealed at 100 °C for 10 minutes. Finally, a 100 nm of Ag electrode

was deposited through a mask by thermal evaporation under a pressure of 5.5×10^{-4} Pa.

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Fabrication of mini-module. Firstly, $5\text{ cm} \times 5\text{ cm}$ FTO glasses were etched using a picosecond laser to form P1 patterns. Prior to use, the etched glasses were cleaned and subjected to ultraviolet-ozone (UVO) treatment. Subsequently, the NiO_x , Me-4PACz, perovskite, PCBM, and BCP layers were sequentially deposited on the patterned FTO glasses in accordance with the fabrication protocols for small-size solar cells. Following the layer deposition, the samples were etched again to form P2 patterns. After evaporating the Cu electrode, a third etching step was performed to create P3 patterns, thereby completing the series connection of 6 subcells. The scribing widths of P1, P2, and P3 were approximately $55\text{ }\mu\text{m}$, $115\text{ }\mu\text{m}$, and $35\text{ }\mu\text{m}$, respectively. The total distance between P1 and P3 was about $210\text{ }\mu\text{m}$, and the calculated geometric fill factor was approximately 96.5%. A $5\text{ cm} \times 5\text{ cm}$ metal mask with an aperture area of $3.6\text{ cm} \times 4\text{ cm}$ was used for the I - V test of these solar mini-modules.

Device characterization. The J - V characteristics were tested using a Keithley 2400 source meter and a Xenon lamp solar simulator (Enlitech, AAA) within an N_2 -filled glove box. The tests were conducted under AM 1.5G irradiation intensity, calibrated with NREL calibrated Si solar cells equipped with infrared cut-off filters (KG-5). Measurements were taken in the range of 1.25 to -0.1 V , with a voltage step of 0.01 V and a delay time of 40 ms .

Device stability was evaluated following the ISOS-L-1 and ISOS-D-2 protocols.³⁹ For the ISOS-L-1 test, unencapsulated devices were continuously exposed to a calibrated white LED light source (100 mW cm^{-2} , AM 1.5G) under ambient conditions ($50 \pm 2\text{ }^\circ\text{C}$) to assess the effect of continuous illumination on device stability. For the ISOS-D-2 test, devices were stored in the dark at an elevated temperature of $85\text{ }^\circ\text{C}$ in ambient air without humidity control to evaluate their thermal stability. No encapsulation was applied during either test, and the device performance was periodically measured using a source meter under standard testing conditions. For the device operating as an LED, the external quantum efficiency was determined by measuring the absolute radiation flux using a commercial system (Guangzhou Xi Pu Optoelectronics Technology Co., Ltd.) equipped with an integrating sphere (GPS-4P-

SL, Labsphere) and a QE-Pro spectrometer (Ocean Optics).

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Other characterizations. Steady-state photoluminescence (PL) was recorded using an FLS1000 instrument (Edinburgh Instruments Ltd). High-performance liquid chromatography (HPLC) measurements were performed using an LC-16P system (Shimadzu). Time-resolved photoluminescence (TRPL) was measured with a 375 nm pulsed laser (EPL-375) excitation using time-correlated single photon counting (TCSPC) equipment (FLS1000, Edinburgh Instruments Ltd). Space-charge-limited current (SCLC) curves were collected using a Keithley 2400 instrument in the dark, with the device's dark J - V curve scanned in the range of -0.5 to 5 V, at a voltage step of 0.01 V and a delay time of 40 ms. Top-view and cross-sectional SEM images were taken using a field-emission scanning electron microscope (JEOL JSM-7610F) operating at 5 kV. X-ray diffraction (XRD) patterns were recorded using a Smart Lab X-ray diffractometer (Rigaku Corporation) with a Cu K α radiation source (1.54 Å), an operating voltage of 40 kV, and a current of 30 mA. Ultraviolet-visible (UV-Vis) absorption spectra were recorded using an Ocean Insight FLAME Miniature Spectrometer in an N₂-filled glovebox. X-ray photoelectron spectra (XPS) were measured using a Thermo Fisher Scientific K-alpha⁺ instrument with an Al K α source, calibrated to the C 1s peak (284.8 eV). Infrared spectra (FTIR) were collected in attenuated total reflection (ATR) mode using a NICOLET iS50 FTIR spectrometer (Thermo Scientific). Incident photon-to-electron conversion efficiency (IPCE) spectra were tested using a QE-R666 system (Enlitech) in DC mode without bias to confirm the current density. The stable output of the device was measured by applying a constant voltage near the maximum power point (MPP) extracted from the J - V curve. Electrochemical impedance spectra (EIS) were obtained using a CHI660E electrochemical workstation, biased at 0.9 V in the dark. Mott-Schottky plots were monitored by varying the voltage from 0 to 1.4 V at 10 kHz using a Zahner electrochemical workstation. Dark J - V curves of the device were recorded using a Keithley 2400 source meter with a scanning range of -1.5 to 1.5 V. Ultraviolet

photoelectron spectroscopy (UPS) spectra were measured using a Thermo Fisher Scientific ESCALAB Xi⁺ with a He-I α (21.22 eV) UV light source. Water-contact angle measurements were performed using a JC2000D1 contact angle measuring device from

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POWEEACH. Time of Flight Secondary Ion Mass Spectrometry (ToF-SIMS) measurements (ION-TOF M6) were performed with the pulsed primary ions from a Cs⁺ (2 keV) liquid-metal ion gun for the sputtering and a Bi⁺ pulsed primary ion beam for the analysis (30 keV). The optical microscope images were recorded on a DM 250 polarized optical microscope.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

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C.T. conceived the idea and supervised the work. E.H. and S.Cheng fabricated devices and analyzed the data. S.K fabricated mini modules. E.H. performed XRD measurements and analyzed the data. S.Chen, Y.X. and S.Cheng performed MALDI-TOF-MS measurements and analyzed the data. Y.Q. performed the DFT calculations and analyzed the theoretical results. J.C. and X.C. contributed to the XPS test. J.Y., L.X. and X.Z. actively contributed to data analysis and discussions. E.H. and S.Cheng co-wrote the manuscript. C.T., Z.W. and T.B revised the manuscript. All authors contributed to the discussion of the data.

Competing interests

The authors declare no competing interests.

Editorial summary:

High cost and aggregation of single-component fullerenes hinder inverted perovskite solar cell development. The authors synthesize a fullerene composite in high yield that cross-links during annealing, improving stability and efficiency while maintaining performance in various device formats.

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